H

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE NAME

CLASS

BIOLOGY

9648/03

Paper 3 Applications and Planning

September 2011

2 hours

Candidates answer questions 1 to 4 on the Question Paper.
Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do no use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Answer questions **1** to **4** in the spaces provided on the question paper.

Answer question **5** on the separate answer paper provided.

For Examiner's Use	
1	
2	
3	
5	
Total	**60**
For Examiner's Use	
4	**12**

This document consists of **13** printed pages and **1** blank page.

[Turn over]

For examiner's use

Section A

Answer **all** the questions in this section.

1 **(a)** Adenosine deaminase (ADA)-deficient SCID is an autosomal recessive genetic disease which affects the immune system in a detrimental manner.

(i) Describe what "autosomal recessive" inheritance of ADA-deficient SCID means.

[2]

(ii) Explain how ADA deficiency results in a SCID.

[3]

(b) Scientists are investigating the use of stem cells in monkeys to treat a CFTR mutation which causes cystic fibrosis in monkeys.

(i) Explain how a mutation in the CFTR gene leads to lung problems in cystic fibrosis sufferers.

[3]

An experiment utilizing genetic engineering techniques was conducted to obtain genetically modified stem cells from monkeys suffering from cystic fibrosis.

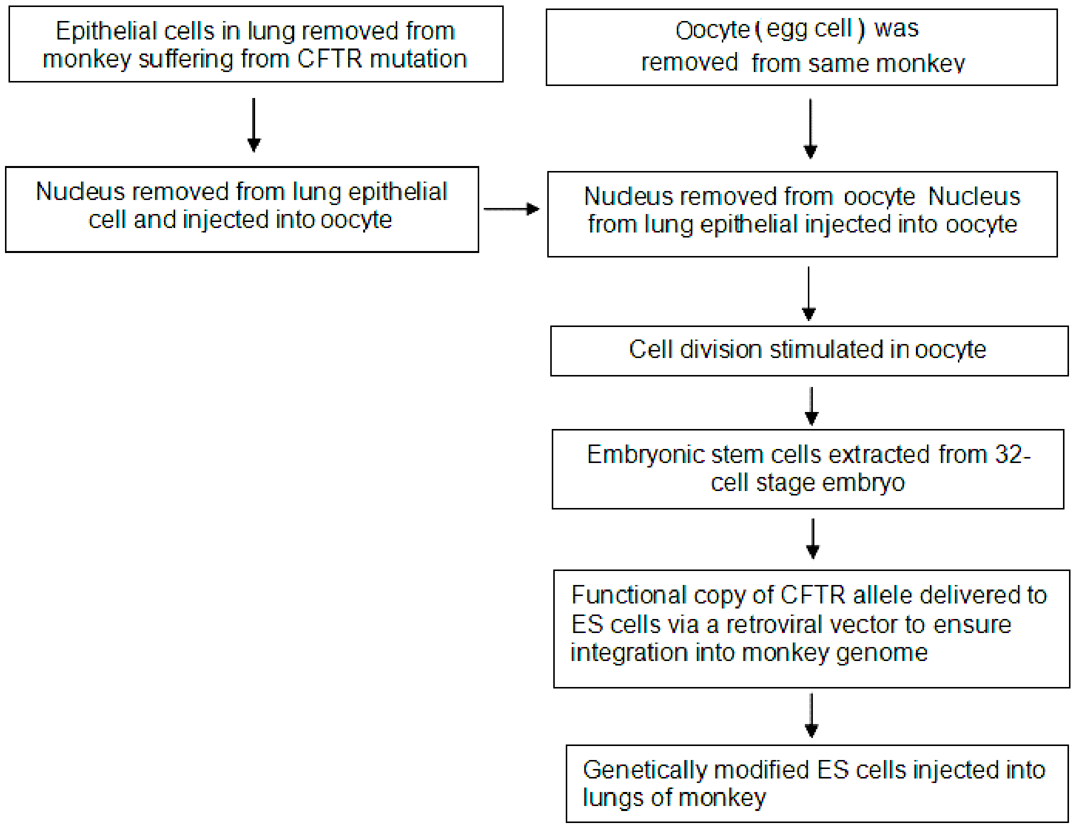

Fig. 1.1

With reference to **Fig. 1.1**,

(ii) describe the characteristics of the ES cells produced.

[2]

(iii) suggest why an oocyte was used and why the nucleus had to be removed from the oocyte.

[2]

[Total: 12]

2 A disease-causing allele is closely linked to an RFLP as shown in **Fig. 2.1**. Two RFLPs are identified for the same loci. **Fig. 2.2** shows a pedigree of a family for the disease and the linkage map for the RFLP locus on a Southern blot.

Fig. 2.1

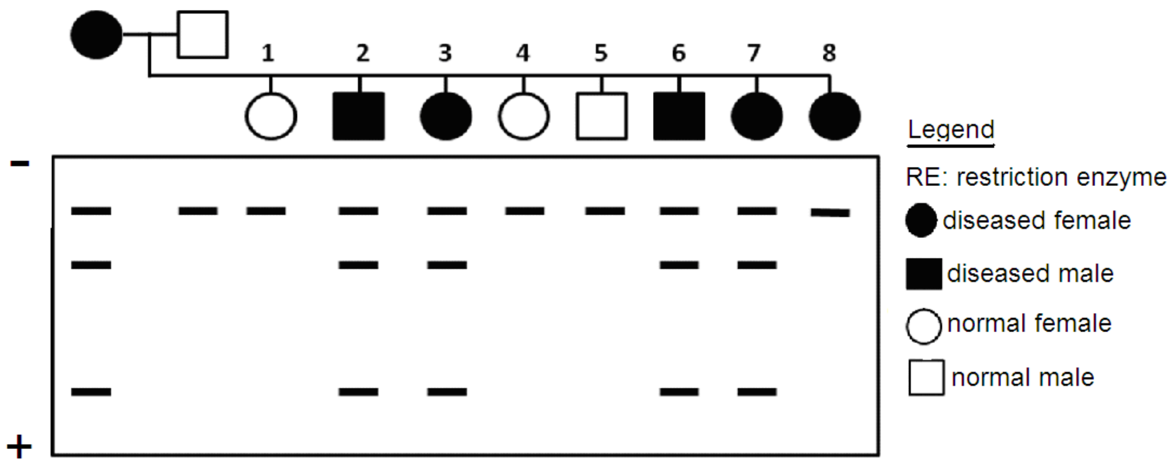

Fig. 2.2

(a) The 2 RFLPs are detected using a probe **P**. On RFLP **1** of **Fig. 2.1**, indicate the position that probe **P** will bind.

[1]

(b) With reference to **Fig 2.1** and **Fig 2.2,**

(i) explain which RFLP the disease-causing allele linked is to;

....

....

[2]

(ii) explain if the disease is caused by a dominant or recessive allele;

....

....

....

....

[3]

(iii) explain how the banding pattern observed in individual 8 arose.

....

....

[2]

For examiner's use

BLANK PAGE

One of the applications of RFLP analysis is for restriction mapping. *Bam*HI, *Eco*RI and *Hind*III restriction enzymes are used to map a 15kb bacterial DNA. Three reactions were set up to cut the 15kb DNA sample, each with one of the three enzymes. After digestion with restriction enzymes, the DNA fragments were separated into bands as shown in **Fig. 2.3**.

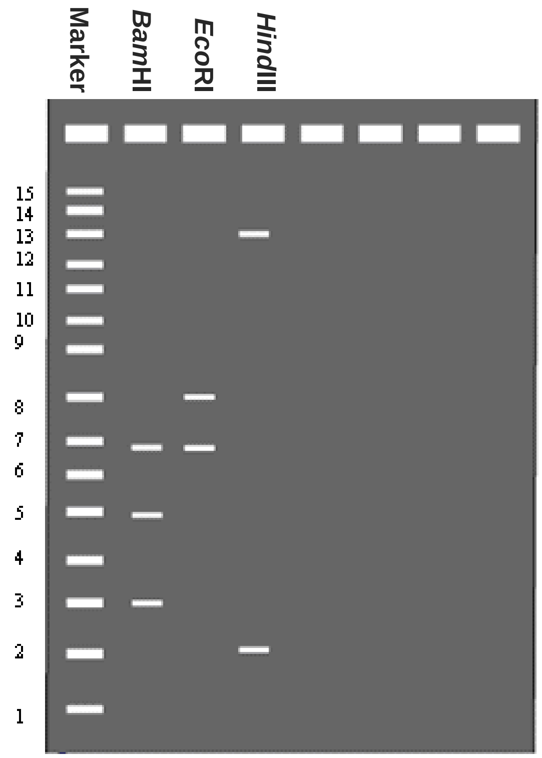

Fig. 2.3

(c) Describe briefly how, in such DNA analysis,

(i) DNA is separated into bands.

....

....

....

[2]

(ii) the bands are made visible.

....

....

[1]

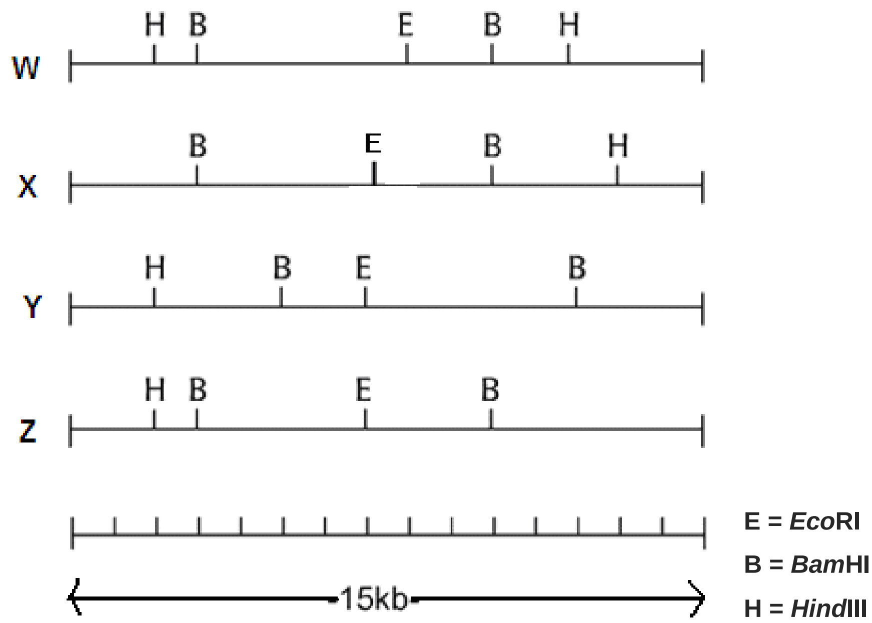

Fig. 2.4

(iii) Based on the results shown in **Fig. 2.3** and **Fig. 2.4**, explain which of the following map (**W**, **X**, **Y** or **Z**) in **Fig. 2.4** is **<u>NOT</u>** a possible map.

....

....

[2]

[Total: 13]

3 The detrimental effect of high salt levels to plants usually cause death or severely stunted growth. A gene encoding a vacuolar Na^+ channel protein from a weed, *Arabidopsis thaliana* was cloned into tomatoes *Lycopersicon esculentum*. Transgenic tomato plants were able to flower and fruit when grown in high salinity conditions.

(a) Suggest the natural role of the vacuolar Na^+ channel protein.

....

....

.... [2]

(b) An experiment was conducted to study the ability to survive on high salinity soil. Twenty plants were used for each condition at the beginning of the experiment. Results of the experiment were shown in **Table 3.1** below.

Type of plant	**Number of plants that survived after transplanting to soil containing NaCl**	
	5 mmol dm^{-3} NaCl	200 mmol dm^{-3} NaCl
Wild type	20	1
Transgenic	20	19

Table 3.1

With reference to **Table 3.1**, describe the survival rate of wild type and transgenic plants in soil containing NaCl

....

....

....

....

.... [3]

For examiner's use

The transgenic plants were grown from tissue culture after successful transformation using a T-plasmid containing a constitutive leaf-specific promoter. Experiments were done to extract the contents of vacuoles of both the wild type and transgenic plants for analysis. The following data was presented.

Fig. 3.2

(c) With reference to **Fig. 3.2**,

(i) describe the data observed in leaves;

....

....

....

[2]

(ii) explain the data observed in leaves;

....

....

....

[2]

(iii) suggest the effectiveness of the constitutive leaf-specific promoter in the expression of the transgene;

....

....

....

....

....

[3]

For examiner's use

(iv) suggest why there were no comparison of wild type plants grown in soil with 200 $mmoldm^{-3}$.

....

[1]

(d) Suggest **two** benefits of this genetically engineered tomato plant.

....

....

....

[2]

[Total: 15]

For examiner's use

4 Skill A – Planning

You are required to plan, but not carry out, an investigation on the difference in duration of each stage in the mitotic cycle, with respect to one another, in an actively dividing cell. This is to establish the longest stage in mitosis.

The mitotic activity of a root tip cell can be estimated by observing cells undergoing cell division with a microscope. When examined under a microscope, the percentage of cells that are undergoing mitosis at different stages can be recorded.

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you should use:

- a microscope
- onion roots with intact tips
- toluidine blue which stains chromosomes blue
- hydrochloric acid at 60°C which fixes the specimen

Your plan should have a clear and helpful structure to include:

- an explanation of the theory to support your practical procedure
- a description of the method used, including the scientific reasoning behind the method and any recommended safety measures
- an explanation of the dependant and independent variables involved
- relevant, clearly labelled diagrams
- how you will record your results and ensure that they are as accurate and reliable as possible
- proposed layout of results tables and graphs with clear headings and labels
- the correct use of technical and scientific terms

[Total: 12]

For examiner's use

For examiner's use

For examiner's use

Free-response question

Write your answers to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)**, etc., as indicated in the question.

5 **(a)** Distinguish between cDNA library and gDNA library. [6]

(b) Describe the aims and benefits of the human genome project. [8]

(c) Outline how plant tissue culture technique is used to produce transgenic plants. [6]

[Total: 20]